

Post-publication custody and cleanup receipt

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Observation: dated snapshot; phases are recorded below

Record: PPC-20260902-02

Repository: sepahead/pid-rs

Machine record: post-publication-custody-2026-09-02.json

Human PDF: post-publication-custody-2026-09-02.pdf

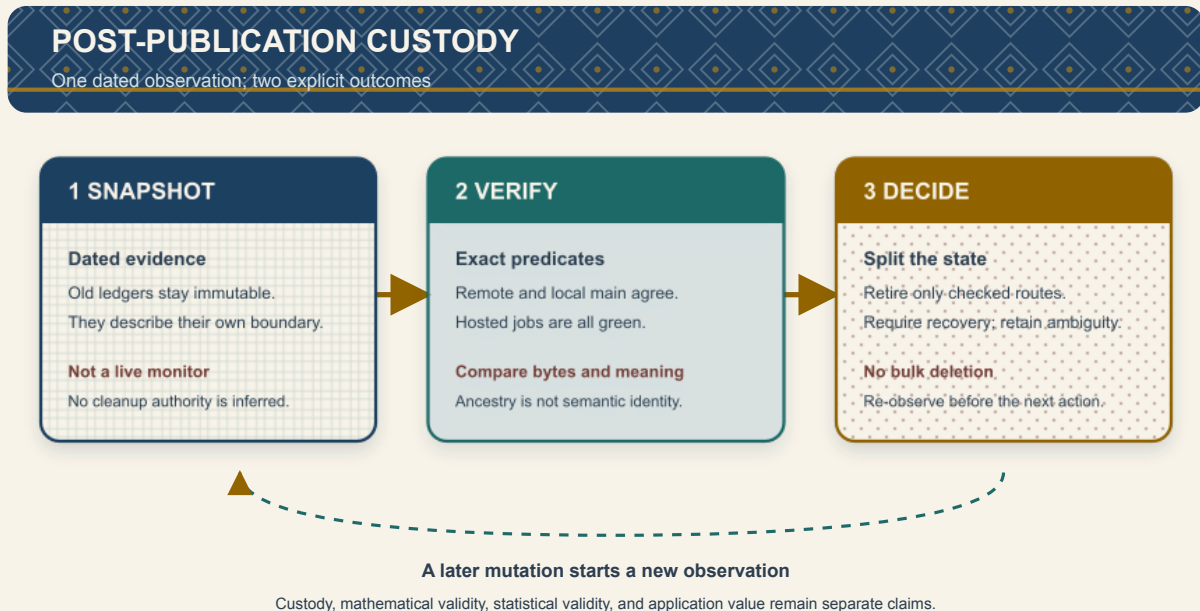
Purpose and boundary

This receipt records a dated observation made after the documentation-closure commit was published and after the named cleanup actions passed their bounded custody predicates. Here, "mainline-published anchor" means the exact commit identified below was reachable from **refs/heads/main** at the recorded observation; it does not mean that every ancestor is mathematically or scientifically accepted. This record is not a live monitor. Re-observe every ref, process, path, and bundle before a later mutation.

The identity below is the snapshot anchor for this receipt. Adding this receipt or later documentation to **main** can create a newer commit; that does not retroactively change the recorded observation. Treat the receipt as dated evidence with an explicit boundary, and create a new observation when the live tip or retained state changes.

The receipt does not rewrite either dated retirement ledger. The primary ledger and the sibling ledger remain immutable, bounded observations made on 1 September 2026 before publication and cleanup. Their fields that say *no cleanup authority* describe those observations. They are not a statement that cleanup never occurred later. This receipt supplies the missing later observation so that a reader does not confuse a historical precondition with the present disposition.

The c499 closure changed documentation, workflow/checker configuration, formal-artifact routing, and repository custody records. This follow-up adds this receipt and its presentation assets. The cleanup operations did not alter Rust/Python estimator code, formal theorem source, test fixtures, or numerical result bytes. Those process changes are bounded and are not scientific validation. No claim here upgrades algebraic formalizations to probability theorems, turns a finite exhaustive check into a population result, or validates an estimator in a new dimension.



The diagram is a process aid. Its colors and ordered patterns follow the repository-local ink, lapis, turquoise, mineral-blue, bronze, and ivory publication palette. The pattern is redundant with the labels, so the state distinction remains readable in grayscale. The SVG source is retained beside the PDF figure for inspection and regeneration. The local figure PDF is a presentation derivative, not a proof artifact.

Observation phases

The first snapshot was opened at 2026-09-02T00:06:18Z. The named cleanup and packed-ref reconciliation actions were recorded complete at 2026-09-02T00:07:35Z; this operator-supplied time is not inferred from a filesystem timestamp. The final local/remote observations occurred later. The machine record keeps separate `observation_started_at_utc`, `actions_completed_at_utc`, and `final_observed_at_utc` fields; it does not silently replace the initial timestamp. The final time is 2026-09-02T04:43:02Z, the timestamp recorded immediately before the bounded pre-follow-up remote-head query and local checks. A later push of this receipt creates a newer `main` tip; it does not rewrite this c499 snapshot.

Machine check and exact remote manifest

The exact remote-head preimage is retained as `post-publication-remote-heads-2026-09-02.tsv`. Each line has one lowercase Git SHA-1 object ID, one TAB, and one `refs/heads/...` name. The file has LF line endings and one final LF. It is a byte preimage of the direct hosted query; it is not a live branch list.

Run this standard-library-only checker from the repository root:

```
python3 -I -S -B scripts/check-post-publication-custody.py
python3 -O -I -S -B scripts/check-post-publication-custody.py
```

The two runs must print the same success line. The checker reads only the JSON record and the TSV; it makes no network request, reads no local Git registry, and performs no mutation. It rejects duplicate JSON keys, non-finite values, private absolute locators, malformed object IDs or ref names, reordered or changed manifest bytes, mismatched digest projections, changed publication identities, altered hosted-run census, weakened cleanup predicates, and missing retention or nonclaim boundaries. The expected snapshot identities are repeated in the checker on purpose. A changed observation therefore requires a reviewed checker update or a new dated receipt, rather than silently changing the meaning of this one.

The `retired_refs_absent_from_observed_manifest` flag has a narrow meaning: the nine named retired ref names are absent from this observed 14-line manifest. It does not prove that a ref can never be recreated, that every copy of its objects is gone, or that a later remote query will be identical. The `remote_post_absence` cleanup note records the separate live query made at the time; this static checker does not repeat that query.

Mainline-published observation anchor

The documentation-closure commit observed on the hosted and local `main` refs was:

- Commit: c499653e4ac89733cb35330bf1a13c93a40ee385.
- Parent: 30e6d19bf020b18ef1cc1f9478c2d4acba62ccf1.
- Tree: 1a1f8dc9782d2f5d6cc9c3342b5395bc7240b975.
- GitHub `refs/heads/main`: c499653e4ac89733cb35330bf1a13c93a40ee385.
- Local primary `refs/heads/main`: c499653e4ac89733cb35330bf1a13c93a40ee385, read from the primary common Git directory with `git -C <primary-repository> rev-parse refs/heads/main`.
- Primary checked-out worktree: `refs/heads/review/sx-count-event-bridge-r2`. Its exact checked-out OID is:
9bbcf5ef04d26b0fd5ec552fe6a065f9a474fd56
- Primary checked-out state: dirty and intentionally retained. It was not overwritten, reset, or merged into by this closure.

The distinction in the last rows is intentional. "Local `main` points to the published commit" does not mean that the currently open primary checkout is clean or is on `main`, and the detached receipt checkout is not evidence for the primary local ref.

After the publication and cleanup checks, the primary common Git directory had stale duplicate values in its packed-ref file: the loose `main` and `origin/main` values were current, but the packed file still named older values. I copied the original packed file to the durable preservation path recorded in the JSON, then ran the narrow Git operation `git pack-refs --all --prune`. The resulting packed file contains `main = c499653e...`, `review/sx-count-event-bridge-r2 = 9bbcf5ef...`, and `origin/main = c499653e...`. The status digest and checked-out OID were identical before and after. This is ref-metadata reconciliation; it is not garbage collection and it does not alter the dirty review files.

A direct `git ls-remote --heads` query observed 14 hosted heads. Its raw-byte SHA-256, including the final newline, is `b8fee7265e8a6ea38adb03324cbc22e07785689e5b55e4a51d2101fce018b82`. The same digest results from the explicit canonical ref-name sort `LC_ALL=C sort -t '$\t' -k2,2` because the server returned the lines in ref-name order. For clarity, the whole-line (OID first) sort has the different digest `1806619eb44aad84806eb63085da1859fa146492611cb72aafefbdac2d3b23c3`. These hashes bind the observed snapshot; the direct query remains the authority for hosted-ref absence. A local remote-tracking cache can still contain names that the host no longer serves, so cached names are not counted as live branches.

The primary clone also contained one stale remote-tracking ref, `refs/remotes/origin/sepahead/pid-rs-release-integration-r4` at `535d7a44e2f8108f806af48cc27b86009239ec4e`. A direct hosted absence check succeeded, the exact old object was verified, and that tracking ref was pruned. The object remains available through the retained archive and bundle routes. This local metadata action does not delete the hosted branch history a second time.

Hosted and local verification

Five required workflow classes ran against the exact publication anchor. Every job completed successfully. The required set is the preregistered push-triggered closure set; the scheduled CodeQL run below is supplementary and is not substituted for a required class.

1. **CI**. Run 33547094635: 47 jobs, success; completed 2026-09-01 23:05:44 UTC.
2. **SxPID3 informative-invariance verification**. Run 33547094668: 3 jobs, success; completed 2026-09-01 19:39:05 UTC.
3. **CodeQL**. Workflow name `CodeQL`, display `Push on main`, event `dynamic`; run 33547093983: 4 jobs, success; completed 2026-09-01 19:11:34 UTC.
4. **Bounded SxPID3 keyed-scalar audit expressions**. Run 33547094598: 1 job, success; completed 2026-09-01 19:11:19 UTC.
5. **KSG M1a composite v12 terminal preservation**. Run 33547094741: 1 job, success; completed 2026-09-01 19:02:01 UTC.

The exact required census is therefore **5/5 runs, 56/56 jobs, zero failures, zero skipped jobs, zero cancellations, zero timeouts, zero action-required jobs, zero neutral jobs, and zero stale attempts**. Here, “stale” is a project-derived count, not a native GitHub field: it means a declared required attempt that was non-completed or superseded at the final observation. A later scheduled CodeQL run, 33558833307, also completed successfully; it is supplementary and is not needed to establish the required census.

The separate throwaway clean c499 validation checkout passed the local `just ci` block and the focused formal, hostile, source-state, method-catalog, publication-link, and PDF checks. The validation checkout was separate from the dirty primary review lane and from the source checkout used to prepare this receipt. The workflow and blueprint PDFs have separate, artifact-specific rendering receipts and 20-lens visual reviews. The closure commit contains no new mathematical implementation.

Remote branch retirement

The following nine remote refs were deleted with exact expected-old-object leases after the published anchor and hosted checks passed. This is a bounded operator action record; the post-action query and route predicates are evidence of the observed state, not an independently witnessed execution transcript:

1. `sepahead/ci-pandoc-toolchain-fix` at `b45a7eb2e15364d37ecffc3061bf4f9ac5812b7f` - superseded by the mainline-published anchor.
2. `sepahead/documentation-closure-v1` at `30e6d19bf020b18ef1cc1f9478c2d4acba62ccf1` - direct predecessor of the mainline-published anchor.
3. `sepahead/galadriel-placement-main-v1` at `eb9c21ae67e7a5cc9279dd7597cc96ed90f062a9` - old main snapshot.
4. `sepahead/openaction-compat-candidate` at `9ed6831d20de43467b1cff8adc8ee421a484f7fd` - superseded compatibility candidate.
5. `sepahead/pdf-annotation-portability-corrected` at `0af14fc97b7c5fe8c4df0361e37cd9cefaa9c6ba` - superseded portability candidate.
6. `sepahead/pid-rs-release-integration-r4` at `535d7a44e2f8108f806af48cc27b86009239ec4e` - integrated release line.
7. `sepahead/pid-rs-release-integration-r4-recovered` at `008ee7fa615aa8370623566c21eb99862680c7b1` - recovered predecessor; bundle custody remains.
8. `archive/composite-v5-unqualified-draft-20260818` at `f7c6122d25ea098a36fa1fc6d672d78f25b783bb` - exact content is contained by the retained rejected-umask archive ref.

9. `sepahead/pid2-rev4-behavior-v1` at `03c0980f256c2a66b3d64bff1686a8d116d76138` - exact content is contained by the retained PID2 assurance ref.

"Contained" means that an exact object and its relevant evidence are reachable through the named retained archive or assurance route. It does not mean that all historical prose is scientifically equivalent. The remote was queried again after the deletion. The remaining archive and diagnostic refs are listed by the live remote, not inferred from an old local cache.

Local worktree and temporary-state retirement

Three primary linked worktrees were removed only after process, cleanliness, ancestry, path, and recovery predicates passed:

1. C4 recovery, former head `bc3aa80fb6025e709c2906a08bce25a4fac40578`. The head is in the published candidate ancestry. A retained same-device bundle observation was 30,416,592 bytes with SHA-256 `503fa4917ab80c801b203c4cb3ee0d0683bb444ba39857eec481cf9a8917ff59`, and `git bundle verify` reported a complete bundle. The private filesystem path is deliberately not reproduced in this public receipt.
2. KSG revision 4, former head `a9aa60c962261a6e0e6698b05551fbcdbf7bf41c`. Its 110 tracked/untracked paths matched the preserved archive head. The historical bundle observation was 11,361,236 bytes with SHA-256 `532ebec0a2a5f2757ccc872925888e1257a082031312d9c7fd4042f6c40cad40`. The raw historical bundle preimage is not present in the current bounded filesystem inventory; this is a recorded historical observation, not a claim of present local retrievability.
3. M1a correction, former head `dc7b8de0a87443ef2bcde71b19938642f1af2197`. Its tracked/untracked paths and ignored-output disposition matched the preserved archive. The historical bundle observation was 11,761,811 bytes with SHA-256 `df4fa378a5a9faf97aa3410ed14a26a0fa870699945b1d91e6b73d2fa72ff2b3`. The raw historical bundle preimage is not present in the current bounded filesystem inventory; this is a recorded historical observation, not a claim of present local retrievability.

The following temporary state was also removed after exact guards:

- separate throwaway clean `c499` validation checkout (not this receipt source checkout);
- clean release-audit checkout at the documented predecessor;
- duplicate render and source-state scratch directories whose hashes matched durable copies or current tracked artifacts;
- the clean `available-audit` clone at `dc7b8de0`, after proving that its refs, reflog objects, and sole unreachable tag were present in the candidate and that its head was an ancestor of `c499`; and
- the selected named throwaway Rust build-target and render-cache paths that were inventoried for this closure. They are rebuild inputs, not scientific records. The primary checkout's `target/`, `tmp/`, and unrelated sibling caches are outside this item and were not globally removed.

The operator record states that no broad repository path was used. Each named target was resolved, checked for links, checked for writers, and removed separately; those details are not an independently witnessed execution transcript. The force option on `git worktree remove` was used only after those checks were reported to show that no required bytes remained; the option itself is not evidence of safety. The recovery bundles and archive refs were not deleted. Git garbage collection was not used as a substitute for an evidence decision. The exact control methods are machine-readable in the `cleanup_controls` object of the JSON record.

State deliberately retained

The cleanup is not a claim that every branch or copy is obsolete. The following state remains protected because the available evidence does not prove redundancy or because another registry still depends on it:

1. The dirty primary review worktree at `9bbcf5ef04d26b0fd5ec552fe6a065f9a474fd56` remains untouched. Its methods narrative, private identifier, unpublished packets, and untracked material require a separate semantic and privacy decision. A fresh private custody package was captured at `2026-09-02T04:36:04Z`. Its exact byte-level inventory is:
 - `v2/NUL` status projection: `991b4d72b2388d05ababf526af40a4eafdf836a3b6a9a2dc0daee269a2407e3a`;

- tracked worktree patch, 1,549,131 bytes: d5b089ef4a5a93d8b96c7ef95d2da6b23c8c08bc75603ced11bdd2216f3537f1;
- empty staged patch: e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855;
- private untracked archive, 1,008,981 bytes: ae94bffad48b7eabae3d6a794fbb7676dbeb6230ea0c63755264e8f5d759ac40;
- all-ref Git bundle, 36,426,597 bytes: 07562762e2a68cffd41f3a44ae76f18fba118802927217f91cf47da5f94259ed;
- bundle verification output, exit status 0: 80aaef8688091f51c56ed399697238506f0e7921f99308da390a1bc475716265; and
- package manifest: a14d4822846bf90c626c0266f283cfaa857abe37705d625647565d0ab395e9c9.

The package captured 159 of 160 untracked leaves after excluding the exact local device-identifier path. These are private byte-preservation facts, not public locators or scientific acceptance.

2. Local `codex/integration-20260804` and `codex/workflow-publication-20260804` remain because each has a divergent publication commit. Their equivalent workflow content is accounted for, but branch removal is not needed for this closure and would erase a useful recovery route.
3. The primary `refs/archive/dirty-primary-pre-integration-20260827`, KSG/M1a archive refs, check-point refs, and quarantine refs remain. They preserve negative results, restricted preimages, and recovery history. A quarantine ref is not an accepted method or release.
4. Remote archive and diagnostic refs remain, including rejected composite experiments, the exact-log hostile route, the PrimeGaps transfer blueprint, the Galadriel guide, PID2 assurance, Python-custody preimage, and C3 diagnostic capture. A side ref is historical evidence, not active `main`.
5. The C12 common registry, its dirty R4 linked worktree, hosted-failure evidence, and L12 evidence remain. They contain nonancestor numerical or custody material that is not reproduced byte-for-byte in the mainline-published anchor.
6. The C11 fresh clone remains because C12 has a configured `c11-source` URL that points to it. Removing it now would leave that URL dangling and break recovery commands until it is rebound.
7. The C3 guide-reproduction clones remain because the divergent capture and two old monolithic SVG blobs have explicit side-ref custody requirements. The current mainline has split replacements, but that is not proof that the old visual history is disposable.

These retained objects are not unfinished accepted work. They are labelled recovery, negative, restricted, or pending-adjudication state. A future retirement must make a new exact comparison and must not infer safety from this receipt.

Presentation artifact and portability boundary

The human-readable PDF is built from this Markdown with `scripts/build-post-publication-custody-pdf.sh`. The builder uses Pandoc 3.10.2, LuaHBTeX 1.18.0 (TeX Live 2024), librsvg `rsvg-convert` 2.62.3, and an explicit Lua filter. The filter projects repository-relative evidence links to `HTTPS://github.com/sepahead/pid-rs/blob/main/...` navigation links in the standalone PDF; it does not change the Markdown source or make the remote target immutable. The machine record binds the builder, header, filter, SVG source, and derived figure hashes after the final build. The PDF is intentionally untagged; no PDF/UA or assistive-technology conformance claim is made. A separate structural check rejects relative/file links and `/GoToR` actions; external URL reachability remains outside this offline check.

Council and lens review

The publication closure was reviewed through the seventy named lens/outcome rows preserved in `PID_DISCOVERY_VERIFICATION_AND_DURABILITY_BLUEPRINT.md`, section “Dated 1 September 2026 adversarial publication closure”: twenty mandatory lenses and fifty additional artifact-specific lenses. The separate PDF receipt records a twenty-lens color/grayscale visual review. This was a self-review council of recorded agents and checks, not independent human or scientific review. The count records process coverage; it is not an assurance probability or a proof of correctness. This custody addendum groups the most relevant questions into the following ten prompts; it does not replace the named seventy-row record:

- Does the mainline-published anchor identify one exact tree, parent, remote ref, and local ref?
- Are the dirty primary checkout and the clean validation checkout kept distinct?

- Does each deletion have a process-free check, an exact target, a reachability or bundle route, and a post-action observation?
- Are branch containment, byte identity, tree identity, ancestry, and semantic equivalence kept as different predicates?
- Are historical ledgers left immutable and clearly dated?
- Are negative, restricted, and failed routes retained without promoting them to `main`?
- Does any wording imply mathematical, statistical, estimator, or application validation that was not performed?
- Are local paths, private identifiers, and temporary build outputs excluded from the public custody claim?
- Can a future agent resume from the JSON record and re-observe the live state?
- Do the Markdown links and PDF references resolve from the repository root?

The disposition is **green with explicit retention caveats for this custody operation**. The council found no source change in this process-only closure that would require altering the PID mathematics; that is not a mathematical endorsement. It also found no basis for an "all branches are gone" claim.

Nonclaims and next observation rule

This receipt does not:

- prove any PID definition, theorem, probability law, estimator consistency, calibration, or downstream sensor-placement benefit;
- authenticate a Git object, bundle, executable, model, or hosted runner;
- turn hosted success into independent scientific review;
- independently witness the operator's deletion commands or prove unrecorded process history;
- make remote-main equality imply a clean working tree;
- make Git reachability, a bundle digest, and a filesystem copy interchangeable;
- provide a global inventory of all disks, sibling directories, hidden processes, or private storage; or
- authorize future ref deletion, worktree removal, cache deletion, or garbage collection.

Before another cleanup action, re-run the remote-ref query, local-ref query, worktree/process inventory, exact status and ignored-path check, bundle verification, and post-action reachability check. If any identity or status differs, stop and create a new receipt. Keep the historical ledgers and this receipt as separate observations. This is the durable process boundary that prevents a new agent from treating a stale snapshot, a branch name, or a polished PDF as authority for a different context.